

Jacob Bollinger

Biomedical Engineer | Medical Device R&D | Product Development | Embedded Systems

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Summary

Biomedical engineer targeting entry-level medical device R&D, product development, systems engineering, and verification/test roles. Experience working with sophisticated healthcare technology in regulated clinical environments, plus hands-on project work in wearable rehabilitation devices, embedded firmware, sensor integration, CAD, 3D printing, motion analysis, and design verification.

Education

Master of Science, Engineering Technology and Management

Concentration in Management of Technology

Sep 2025 – Expected Mar 2027

Louisiana Tech University

Bachelor of Science, Biomedical Engineering

Concentration in Mechanical Engineering

Sep 2020 – May 2024

Louisiana Tech University

Technical Skills

Medical Device Development: Design verification, validation, testing, technical documentation, risk management concepts, human factors, regulated healthcare environments

Systems & Product Development: Product development, systems integration, hardware/software interaction, root cause analysis, troubleshooting, cross-functional collaboration

Embedded Systems: ESP32, Arduino, BNO055 IMU, accelerometers, sensor integration, Bluetooth Low Energy, firmware, motion analysis, repetition counting, exercise-state tracking

Design & Prototyping: SolidWorks, CAD, 3D printing, enclosure concepts, hardware development, rapid prototyping, human factors, component packaging

Analysis & Software: Python, MATLAB, COMSOL, Excel, data analysis, Salesforce

Experience

Support Specialist

Brainlab Inc.

Jun 2024 – Present

Dallas, TX

- Install, configure, calibrate, and verify advanced image-guided radiation therapy systems in clinical environments, supporting deployment of complex medical technologies used in patient treatment.
- Perform root-cause analysis across hardware, software, imaging, and networked subsystems to diagnose technical issues and restore system performance in regulated healthcare environments.
- Execute system calibration, verification, and validation activities to ensure compliance with manufacturer specifications and clinical performance requirements.
- Collaborate with physicians, medical physicists, dosimetrists, and clinical staff during system commissioning, acceptance testing, and workflow integration.
- Analyze interactions between imaging, tracking, treatment-planning, and software components to identify failure modes and implement corrective actions.
- Create technical documentation, service reports, and issue-resolution records while maintaining traceability of maintenance and corrective actions.
- Develop expertise in high-precision medical technologies, healthcare workflows, and cross-functional problem solving within safety-critical clinical environments.

Earlier Professional Experience: Construction Laborer, Markovich Construction LLC

Jun 2023 – Aug 2023

Earlier Professional Experience: Caddie, Whistling Straits

May 2022 – Aug 2022

Engineering Projects

Medical Device Development Projects

Wearable Rehabilitation Motion Tracker

Embedded Medical Device Prototype

Technologies: ESP32, BNO055 IMU, Bluetooth Low Energy (BLE), Embedded C++, Arduino IDE, Sensor Integration

- Designed and developed a wearable rehabilitation tracking device to monitor patient movement, count repetitions, and evaluate exercise quality during rehabilitation exercises.

- Integrated ESP32 microcontroller, BNO055 inertial measurement unit, Bluetooth Low Energy communication, rechargeable power system, and wearable enclosure architecture into a functional prototype.
- Developed embedded firmware for sensor acquisition, exercise-state detection, repetition counting, movement analysis, and real-time device operation.
- Implemented motion-analysis algorithms to identify exercise phases, track movement depth, and support future exercise-form assessment capabilities.
- Created enclosure concepts and system architecture for wearable deployment, balancing sensor placement, usability, charging accessibility, and component packaging constraints.
- Applied medical-device development principles including human factors, iterative prototyping, verification planning, design refinement, and system-level problem solving.

OsteoX Surgical Bone Removal Device

Senior Design Project

- Developed a powered neurosurgical bone-removal device concept intended to improve procedural efficiency while protecting surrounding tissue.
- Designed a compact works-like device in CAD, including handle, shaft, milling-bit assembly, motor package, sensor path, and internal component layout.
- Fabricated and evaluated multiple 3D-printed prototypes to assess ergonomics, maneuverability, usability, and design feasibility.
- Evaluated design tradeoffs involving sterilization requirements, manufacturability, procedural workflow integration, and surgical usability considerations.
- Conducted iterative design improvements based on prototype testing, component packaging constraints, and performance objectives.

Certifications & Courses

Certified SOLIDWORKS Associate — Supervised Machine Learning: Regression and Classification — Programming for Everybody: Getting Started with Python — Python Data Structures — Using Python to Access Web Data